

IE598 Machine Learning in Fin Lab

Module 4 Homework (Regression)

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Introduction

This report analyzes the Boston Housing dataset, which includes 13 additional random attributes, using linear regression, Ridge regression, and LASSO regression. We first explore the dataset through descriptive statistics and visualizations. We then fit and evaluate the three regression models using mean squared error (MSE) and R^2 , and compare their predictive performance.

1 Exploratory Data Analysis

1.1 Dataset Description

The dataset used in this analysis is `housing2.csv`, an extended version of the Boston Housing dataset. It contains 506 observations and 27 variables. The original dataset includes 13 explanatory variables describing characteristics of housing and neighborhoods in the Boston metropolitan area, while the extended dataset additionally contains 13 randomly generated variables, `ATT1–ATT13`. The target variable is `MEDV`, which represents the median value of owner-occupied homes.

The explanatory variables include both continuous variables and the binary variable `CHAS`. The remaining variables are numerical. There are no missing values in the dataset. The data are therefore complete and can be used directly for the regression analysis.

1.2 Data Visualizations

The distribution of the target variable, `MEDV`, is shown in Figure 1. The distribution is concentrated around approximately 15–25. The large spike at the upper limit of 50 suggests that some observations may have been capped at this upper threshold.

The correlation structure among the housing variables is shown in Figure 2. The heatmap provides an overview of the linear relationships between the predictors and `MEDV`, as well as potential correlations among predictors. Several of the original housing variables exhibit relatively strong correlations with each other and with the target variable, while the additional variables `ATT1–ATT13` generally show weaker correlations with the other variables and with `MEDV`.

Figure 3 presents the pairwise relationships among the original Boston Housing variables, including `MEDV`. Figure 4 presents the corresponding relationships for the additional variables `ATT1–ATT13`. Due to the number of variables, the pair plots provide limited detail, but the original variables show more varied distributions and apparent relationships, whereas the additional variables appear to be more evenly distributed.

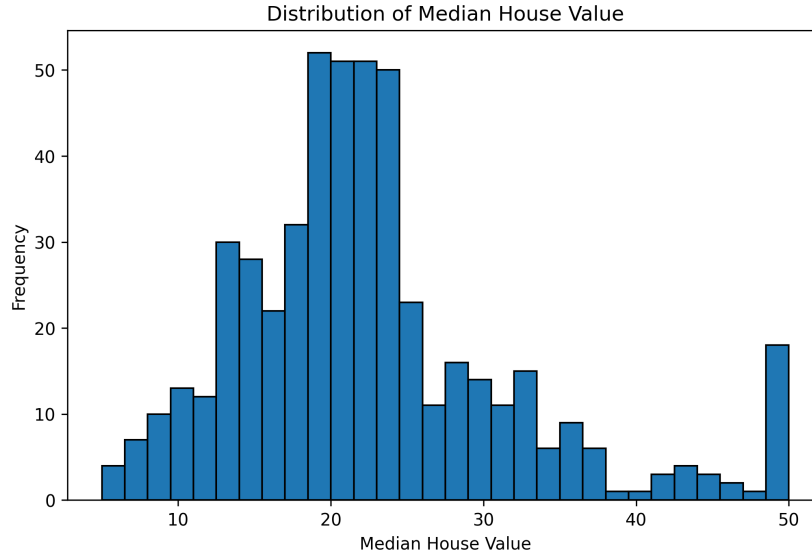


Figure 1: Distribution of Median House Value (MEDV)

2 Linear Regression

Linear regression was first fitted as a baseline model for comparison with Ridge and LASSO regression. The fitted linear regression model can be expressed as

$$\hat{y} = \beta_0 + \mathbf{x}^\top \boldsymbol{\beta},$$

where β_0 is the intercept, $\mathbf{x}$ is the feature vector, and $\boldsymbol{\beta}$ is the coefficient vector. The model is fitted using all 26 standardized features.

2.1 Model Coefficients

The fitted model has an intercept of 22.7965. Since the features were standardized, the coefficients represent the changes in the predicted target associated with a one-standard-deviation increase in each feature, holding other features constant. Among the original variables, LSTAT (-3.5584), RM (3.1756), and DIS (-3.1299) have relatively large coefficient magnitudes.

2.2 Performance and Residual Analysis

On the test set, the linear regression model achieves an MSE of 26.6302 and an R^2 of 0.6369. The residual plot is shown in Figure 5. The results are compared with those of the Ridge and LASSO models in the final section.

3 Ridge Regression

Ridge regression was used to reduce the impact of potentially large or unstable coefficients, particularly those associated with the additional attributes. Ridge regression adds an L2 penalty to the loss function, shrinking the coefficients toward zero without forcing them to become exactly zero. The regularization strength is controlled by the parameter α . We tested multiple values of α ranging from 10^{-3} to 10^2 using five-fold cross-validation, and selected the value with the lowest validation MSE to fit the final Ridge model.

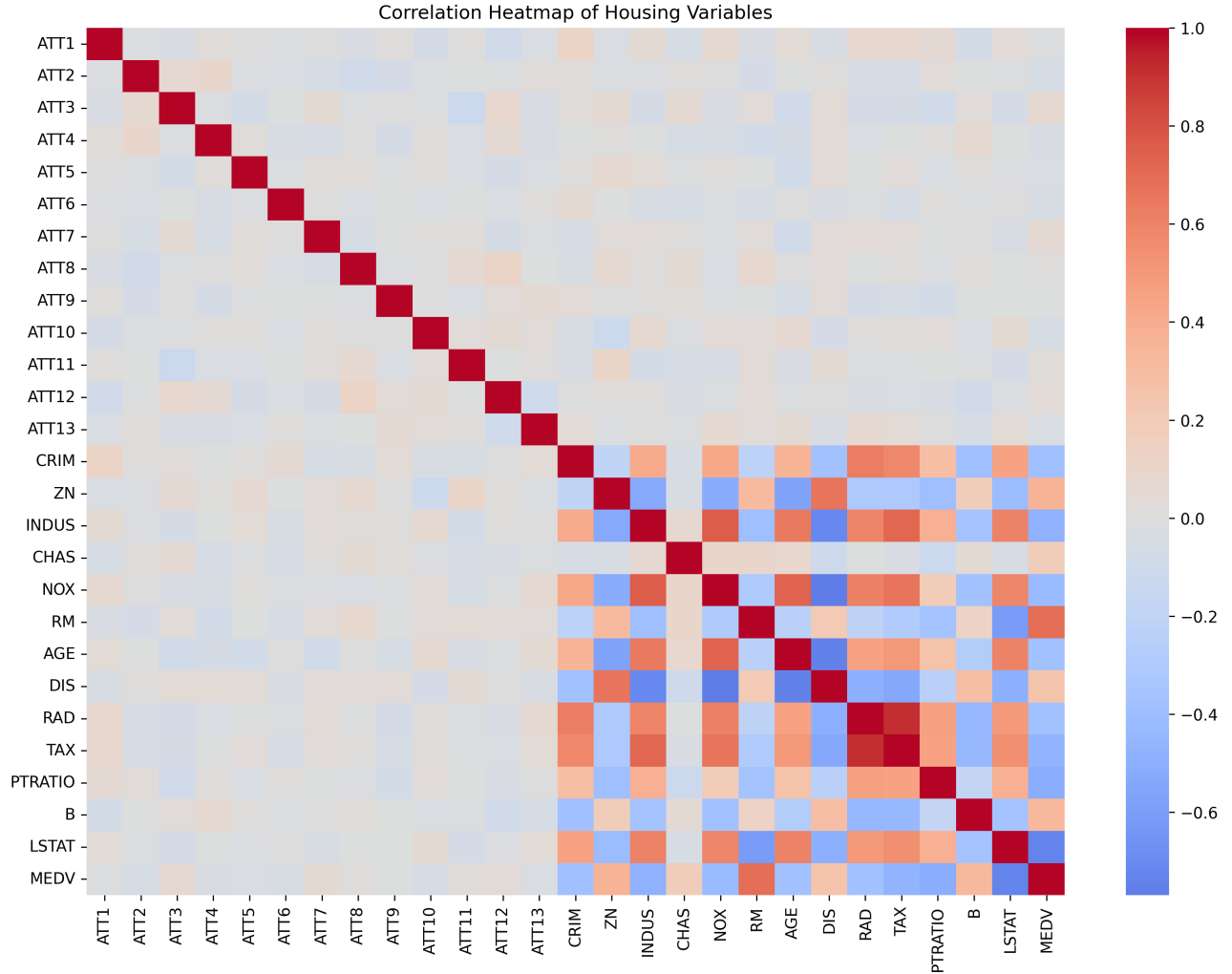


Figure 2: Correlation Heatmap of Housing Variables

3.1 Model Selection and Coefficients

Figure 6 shows the cross-validation MSE for different values of α . The lowest cross-validation MSE was obtained at $\alpha = 9.2367$, with a cross-validation MSE of 25.1630. This value was selected for the final Ridge model.

The fitted Ridge model can be expressed in vector form as

$$\hat{y} = \beta_0 + \mathbf{x}^T \boldsymbol{\beta},$$

where $\mathbf{x}$ is the 26 standardized features and $\boldsymbol{\beta}$ is the coefficient vector. The fitted model has an intercept of $\beta_0 = 22.7965$.

The coefficient paths are shown in Figure 7, illustrating how the coefficients change as α increases. Compared with linear regression, Ridge regression shrinks the coefficients toward zero as α increases. Although we expected the randomly generated attributes to shrink toward zero more quickly, this pattern is not particularly pronounced in the figure because their coefficients are already small and decrease only slightly relative to those of the original features.

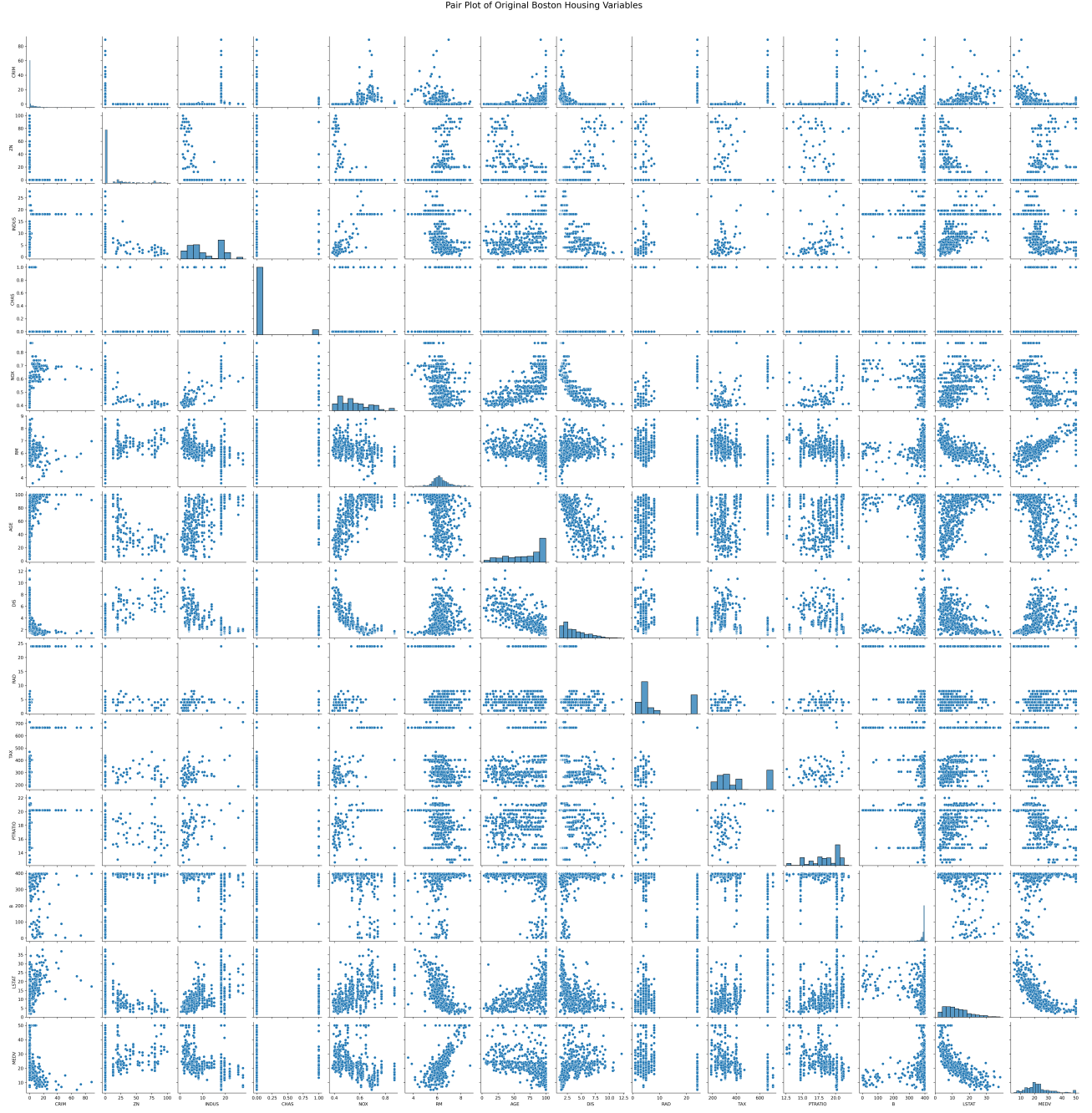


Figure 3: Pair Plot of Original Boston Housing Variables

3.2 Performance and Residual Analysis

On the test set, the best Ridge model achieves an MSE of 26.7404 and an R^2 of 0.6354. The residual plot is shown in Figure 8. The performance of Ridge regression is compared with that of linear regression and LASSO regression in the final section.

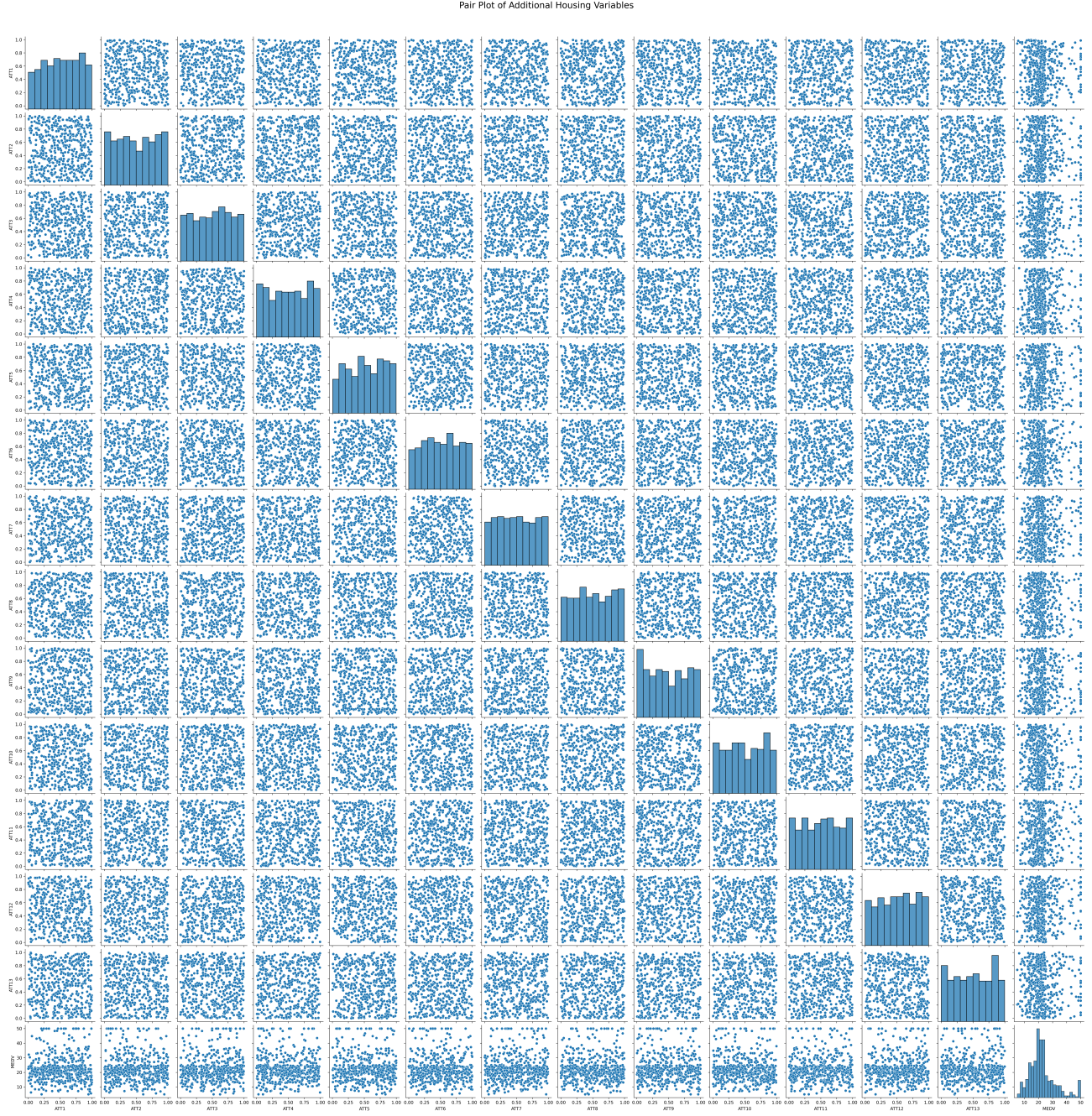


Figure 4: Pair Plot of Additional Housing Variables

4 LASSO Regression

LASSO regression was used to reduce the impact of potentially less important features. LASSO adds an L1 penalty to the loss function, which can shrink some coefficients exactly to zero and therefore perform feature selection. The regularization strength is controlled by the parameter α . We tested multiple values of α ranging from 10^{-3} to 10^2 using five-fold cross-validation, and selected the value with the lowest validation MSE to fit the final LASSO model.

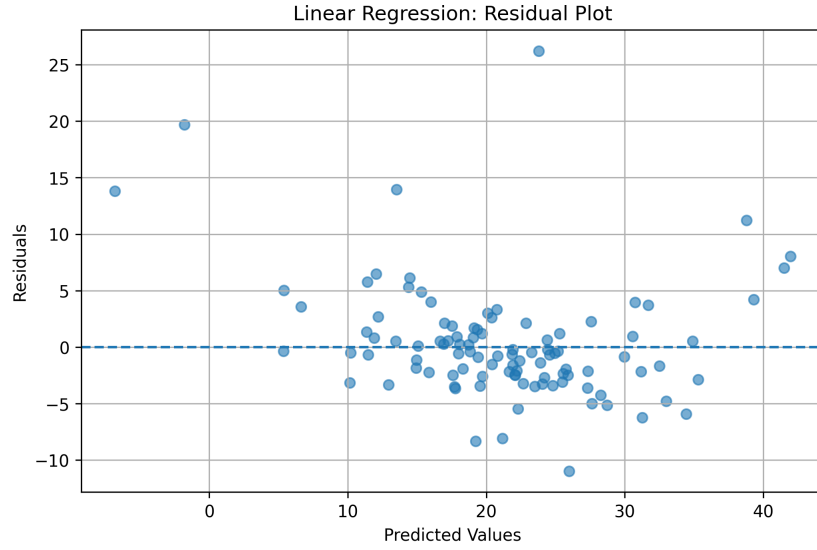


Figure 5: Residual Errors for Linear Regression

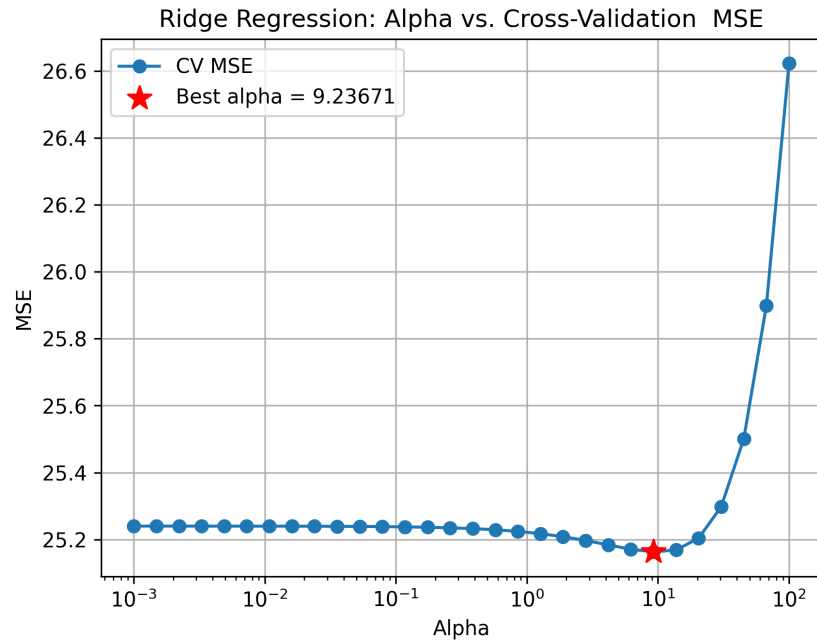


Figure 6: Ridge Regression: Cross-Validation MSE

4.1 Model Selection and Coefficients

Figure 9 shows the cross-validation MSE for different values of α . The lowest cross-validation MSE was obtained at $\alpha = 0.1743$, with a cross-validation MSE of 24.9336. This value was selected for the final LASSO model.

The fitted LASSO model can be expressed in vector form as

$$\hat{y} = \beta_0 + \mathbf{x}^\top \boldsymbol{\beta},$$

where $\mathbf{x}$ contains the 26 standardized features and $\boldsymbol{\beta}$ is the coefficient vector. The fitted model has an intercept of 22.7965.

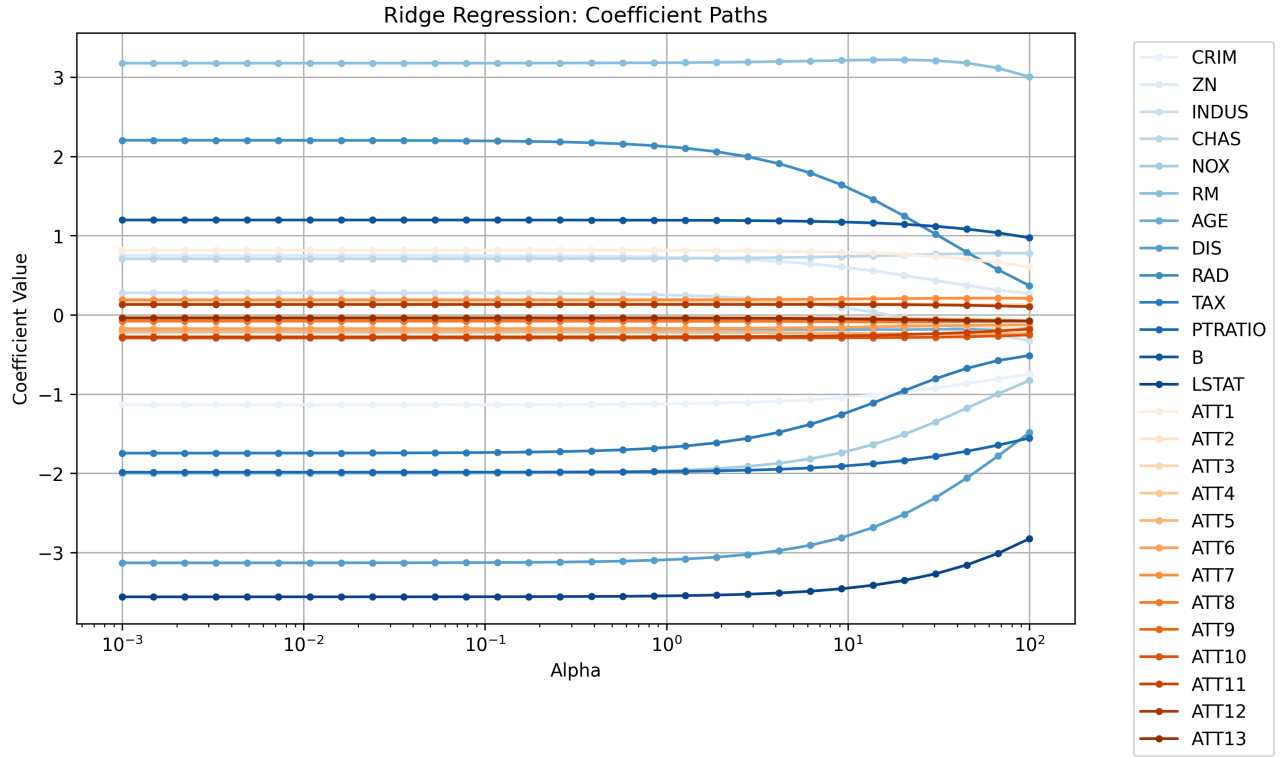


Figure 7: Ridge Regression Coefficient Paths

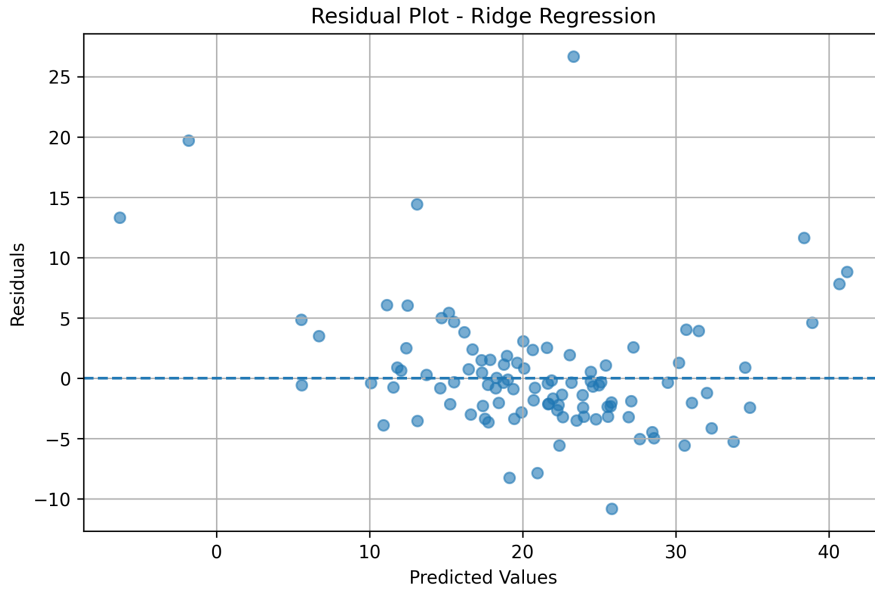


Figure 8: Residual Errors for Ridge Regression

The coefficient paths are shown in Figure 10, illustrating how the coefficients change as α increases. Unlike Ridge regression, LASSO can shrink coefficients exactly to zero. At the selected α , 12 of the 26 coefficients are zero, including 8 of the 13 additional attributes. Among the original variables, INDUS, AGE, RAD, and TAX are also removed from the model.

4.2 Performance and Residual Analysis

On the test set, the best LASSO model achieves an MSE of 28.9058 and an R^2 of 0.6058. The residual plot is shown in Figure 11. The performance of LASSO regression is compared with that of linear regression and Ridge regression in the final section.

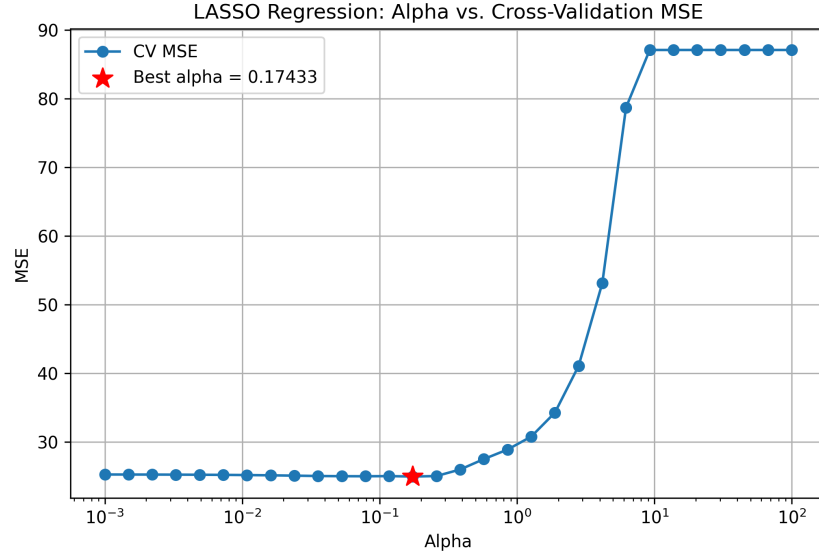


Figure 9: LASSO Regression: Cross-Validation MSE

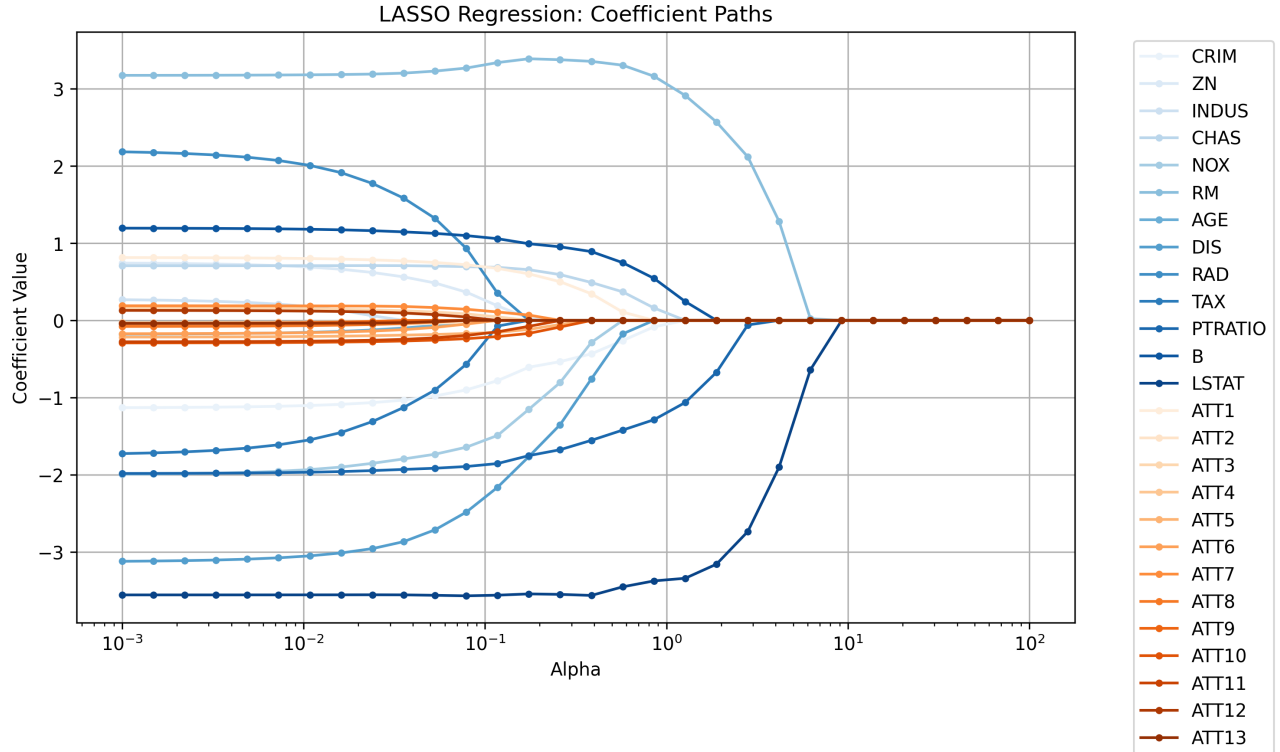


Figure 10: LASSO Regression Coefficient Paths

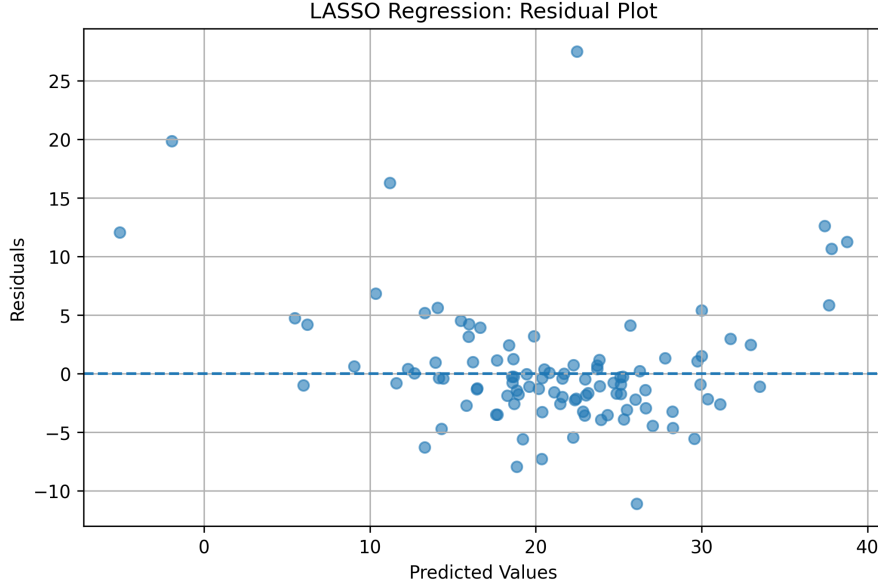


Figure 11: Residual Errors for LASSO Regression

5 Conclusions

The test-set performance of the three regression models is summarized in Table 1. Linear regression and Ridge regression show very similar predictive performance, with Ridge producing slightly higher MSE and slightly lower R^2 .

Table 1: Test-Set Performance Comparison		
Model	MSE	R^2
Linear Regression	26.6302	0.6369
Ridge Regression ($\alpha = 9.2367$)	26.7404	0.6354
LASSO Regression ($\alpha = 0.1743$)	28.9058	0.6058

In addition to regularization, LASSO performed feature selection by setting 12 of the 26 coefficients to zero, including 8 of the 13 additional attributes. However, several additional attributes remained in the model, LASSO did not remove all of the randomly generated variables. On the other hand, Ridge regularization slightly shrank the coefficients but did not substantially improve the test-set performance.

The residual plots show similar overall patterns across the three models, suggesting that regularization did not substantially change the residual behavior. This result suggests that randomly generated attributes can exhibit some correlation with the target in the dataset due to random variation, and regularization may not be able to eliminate all such variables from the fitted model.

A GitHub Repository

The code and analysis for this report are available in the GitHub repository:

https://github.com/ronggan01/Machine-Learning-in-Finance-Fall-2026/tree/main/Boston_Housing_Regression_Analysis